

No. of Printed Pages : 4
Roll No.

221064B

**6th Sem/
ECE, ECE (For Speech & Hearing Impaired)
Subject : Medical Electronics**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Strain gauge is used to measure
- a) Temperature b) Pressure
 - c) Height d) Displacement
- Q.2 What is the normal pulse rate of Human being?
- a) 20-40 b) 60-100
 - c) 100-150 d) above 150
- Q.3 LVDT works on the principle of
- a) Variable resistance b) Variable inductance
 - c) Variable capacitance d) Variable pressure
- Q.4 Primary function of a pace-maker is :
- a) To regulate blood pressure
 - b) To monitor cholesterol levels
 - c) To maintain normal heart rhythm
 - d) To pump blood to the body

- Q.5 Source of Bio electric potential is _____ in nature.
- a) Electronic b) Electric
c) Ionic d) Mechanical
- Q.6 _____ converts bio chemical events into measurable signals.
- a) Amplifier b) Op-amp
c) Rectifier d) Transducer

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Heart Rate.
- Q.8 Define Defibrillator.
- Q.9 What is the full form of MRI?
- Q.10 Define Bio Sensors.
- Q.11 What is leakage current?
- Q.12 What is Micro current shock?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Draw and Explain surface electrode.
- Q.14 Explain Safety Standards.
- Q.15 Explain any one type of temperature transducer.

- Q.16 Write a note on biological amplifiers.
- Q.17 What is Electrode Tissue Interface?
- Q.18 Explain the ultrasonic imaging system?
- Q.19 Write a note on photoelectric transducer.
- Q.20 Write the need of defibrillator. Write a note on photoelectric transducer.
- Q.21 Define Bio-Electric signal.
- Q.22 List any four clinical laboratory - equipments.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Draw Block diagram and Explain working of EEG machine.
- Q.24 Give the classification of bio-transducers. Explain the pulse sensor.
- Q.25 Write short note on :
- a) Bedside Patient monitoring system.
 - b) Heart rate Measurement

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